

Mohamed Ahmed El-Ekhtiar

AEC Software Developer

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Professional Summary

AEC Software Developer with a background in Electrical Engineering, specializing in BIM automation and .NET development. Skilled in C#, Object-Oriented Programming, software architecture, and Revit API development. Passionate about developing automation tools that streamline BIM workflows, improve productivity, and support engineering design processes. Strong interest in BIM technologies, geometric algorithms, computer graphics, and 3D visualization, with a commitment to continuous learning and building innovative software solutions for the AEC industry.

Education

Information Technology Institute (ITI) <i>9-Month Professional Diploma - AEC (Architecture, Engineering, and Construction) Informatics Track</i>	Oct 2025 – In Progress <i>Smart Village, Egypt</i>
Mansoura University <i>Bachelor of Electrical Power and Machines Engineering</i>	2016 – 2023 <i>Mansoura, Egypt</i>
Completed graduation project on designing a stand-alone photovoltaic (PV) system, applying electrical design fundamentals, load estimation, and PV components sizing principles.	

Projects

Electrical BIM Automation Revit Add-in <i>C#, WPF, Revit API</i>	- Building a computational lighting design and calculation engine using C# and .NET to automate optimal luminaire placement, lighting calculation and spatial room analysis. - Developing a photometric data extractor to parse IES files, dynamically rendering polar intensity plots and real-time Lux visualization heatmaps via Revit's AVF. - Programming a shop drawing automation algorithm to instantly place lighting back boxes, route conduits, and generate layout dimensions, circuit tags, and documentation sheets.
Galaxy Scene Simulation <i>C++, OpenGL</i>	- Developed an interactive 3D galaxy simulation using modern OpenGL, shaders, textures, and real-time rendering techniques. - Implemented camera movement, object transformations, hierarchical modeling, and planetary orbital motion using transformation matrices.
E-Commerce Web Application <i>ASP.NET MVC, Entity Framework, SQL Server</i>	- Built a web-based e-commerce application with product, customer, and order-management workflows. - Implemented MVC architecture, Entity Framework models, CRUD operations, and relational database design using SQL Server. - Implemented a secure Authentication and Authorization system using ASP.NET Identity, enabling multi-tier role management (Admin, Vendor, Customer) and securing sensitive checkout and dashboard routes.
Construction Material Manager <i>C#, WPF, MVVM</i>	- Developed a desktop application for managing construction materials with CRUD operations and organized data-management workflows. - Applied MVVM architecture, data binding, and commands to create a maintainable WPF interface.

Courses & Training

Electrical Distribution Diploma – Eng. Mohamed Emam <i>Electrical Design Course</i> - Completed an electrical design project for a call center, including lighting design using Dialux Evo, small power systems, and panel schedules. - Performed cable selection, SLD preparation, transformer, generator, and UPS sizing for building electrical systems. - Designed earthing, lightning protection, street lighting, cable routing, and basic infrastructure systems.	Jul 2022 – Oct 2022
Misr Spinning & Weaving Company <i>Summer Trainee - Power Station</i> - Observed power station operations and electrical distribution workflows, connecting academic power systems knowledge with industrial maintenance and operation practices. - Analyzed control, protection, and equipment-monitoring procedures to build practical awareness of reliability, safety, and troubleshooting in large-scale facilities.	Aug 2021 – Sep 2021

Technical Skills

Programming & Software Development: C, C#, C++,JavaScript, OOP, Data Structures, .NET, WPF (MVVM), LINQ. Databases & Web Development: SQL Server, ADO.NET, Entity Framework, ASP.NET MVC, HTML5, CSS3, JavaScript. AEC & BIM Technologies: AutoCAD ,Revit,BIM, Revit API, Dynamo, Autodesk Revit MEP. GIS & Mapping: ArcGIS Pro, ArcGIS Online. 3D & Visualization: OpenGL,Three.js. Tools: Git/GitHub.

Languages

Arabic	Native
English	Very Good

Additional Information

Military Status	Completed
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